

YOSR BEN NAGRA

AI/ML Engineer

yosrbennagra@gmail.com | +216 53916040 | Tunis, Tunisia | portfolio-yosr.vercel.app | github.com/YosrBennagra | linkedin.com/in/yosr-ben-nagra

Professional Summary

AI/ML Engineer with hands-on experience building end-to-end ML systems. Full-stack developer transitioning into machine learning with solid foundations in **Python**, **data processing**, and **model deployment in production**. Completed PFE developing a complete AI medical diagnosis platform (React frontend + Flask backend + RAG + Transformers). Eager to deepen expertise in ML engineering, model optimization, and data pipelines. Strong analytical mindset and quick learner.

Education

ESPRIT, Computer Science Engineering (Web and Internet Technology specialization), 2020–2025, Tunis

Lycee Soukra, CS Baccalaureate, 2019

Technical Skills

Python & Data: Python (pandas, numpy); working knowledge of scikit-learn; Hugging Face Transformers (fine-tuning), PyTorch basics

AI/LLM Exposure: Generative AI concepts, RAG pipeline design, prompt engineering, OpenAI API integration

Data Processing: Data cleaning, preprocessing, EDA (pandas, seaborn, matplotlib); working with medical datasets

Backend & APIs: Flask, FastAPI; REST API design and integration

DevOps & Deployment: Docker, basic CI/CD (Jenkins), model containerization and deployment

Databases: PostgreSQL, MongoDB, SQL queries, data pipeline basics

Web Integration: React, TypeScript (for dashboard/UI components)

Soft Skills: Problem-Solving, Continuous Learning, Analytical Thinking, Communication, Adaptability, Team Collaboration, Autonomous

Professional Experience

AI/ML Engineer (PFE) – Medical Diagnosis Platform

Itserv, Tunis

02/2025 – 08/2025

- Built complete end-to-end AI system: **React/TypeScript** frontend, **Flask** REST API backend, ML pipeline
- Data pipeline: collected medical datasets, performed cleaning/preprocessing, basic EDA using pandas/matplotlib
- Fine-tuned Transformer model** (Hugging Face) on medical data for disease prediction
- Implemented **RAG pipeline** for context-aware inference; integrated with REST API
- Deployed application using **Docker** and basic **CI/CD** (Jenkins); set up monitoring with Prometheus/Grafana
- Worked with hybrid database (PostgreSQL + MongoDB); learned model versioning and deployment practices

Full Stack Developer – Collaborative Documents Platform

ESPRIT, Tunis (Academic Project)

2024

- Developed backend with real-time data processing and API optimization
- Implemented data serialization and caching for high-throughput applications

Backend Developer – Educational Platform

Ironbyte, Tunis

2024

- Built **Node.js/Express** backend with optimized database schemas (indexing, query tuning) for efficient data retrieval
- Implemented data aggregation pipelines with filtering, pagination, and full-text search

Key Projects

Medical AI Diagnosis System (PFE)

Full Stack + ML

Python, React, Flask, Hugging Face

- Integrated frontend (React dashboard) + backend (Flask API) + ML model (fine-tuned Transformer)
- Learned data pipeline concepts: collection, cleaning, preprocessing, EDA
- Implemented RAG system for retrieval-augmented diagnosis support
- Deployed with Docker; basic CI/CD and monitoring setup

Certifications

MongoDB – Building RAG Apps, 2025

Hadera – Hashgraph Developer, 2025

Saylor Academy – C++, 2022

Languages

English – Fluent

French – Professional

Arabic – Native